

MF20006: Introduction to Computer Science

Lecture 3: Computer Networks

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Outline

1. Information Transmission

2. IP and Routing

3. Messaging

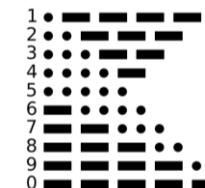
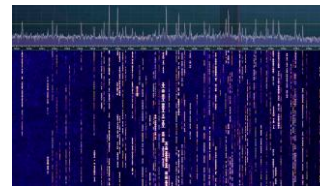
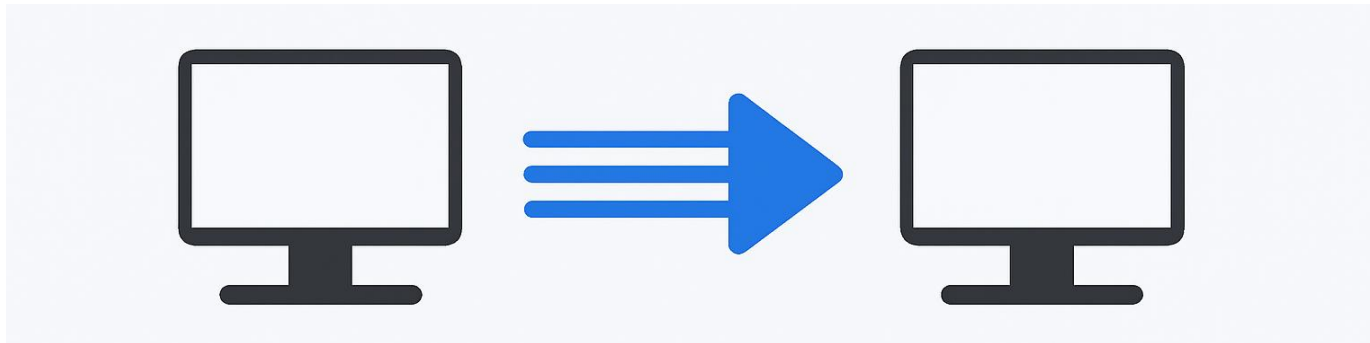
4. Practice: Packet Analysis

Scenario: Direct Communication Between Two Endpoints

❑ How to **transmit** information between two endpoints?

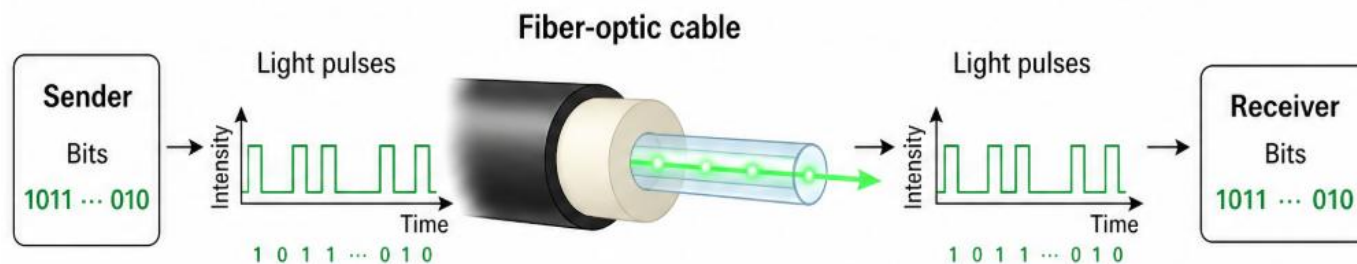
❑ What kinds of information?

- Morse code? Human-readable, but not required.
- Bits are enough for computers. Leave the interpretation to upper layers.



Transmission via Wired Communication

- ❑ Use physical media such as copper cables or optical fibers.
- ❑ Send data (bits) as electrical or light signals over the wire.
 - Electrical signals: the bits are turned into tiny changes in voltage or current.
 - Light signals: In a fiber-optic cable, bits are represented by pulses of light.
- ❑ The data is encoded or modulated onto the physical signal and recovered at the receiver.



Transmission via Wireless Communication

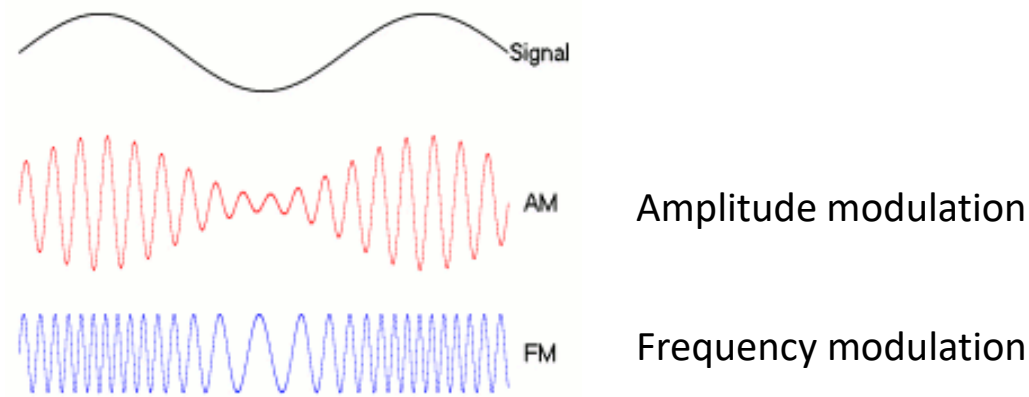
- ❑ Send information using electromagnetic waves.

 - Examples: Wi-Fi, Bluetooth, Cellular (4G/5G), Satellite.

- ❑ Digital data is modulated onto a carrier wave.

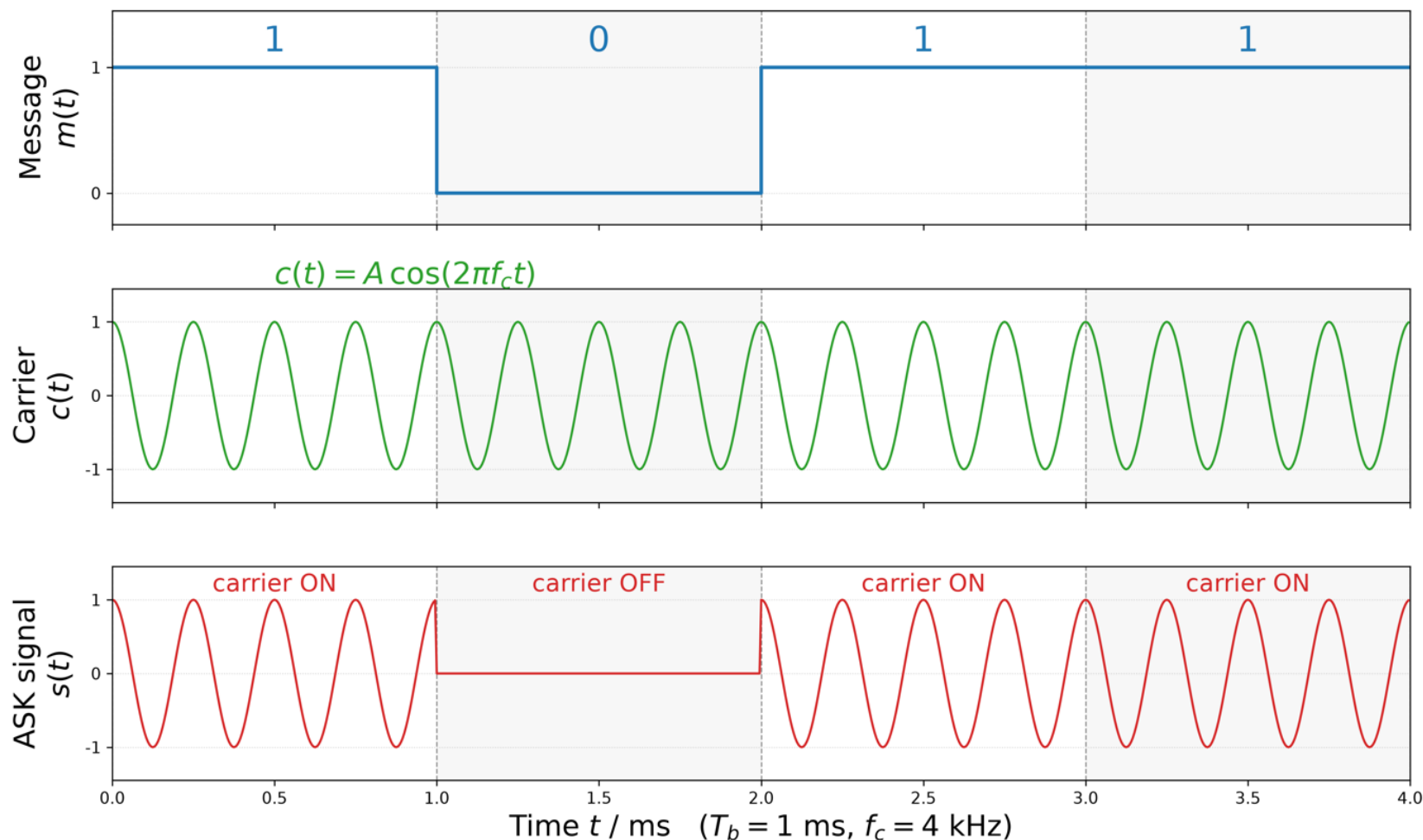
- ❑ Receiver demodulates the signal to recover the original data.

- ❑ Methods: amplitude/frequency modulation



Amplitude Modulation

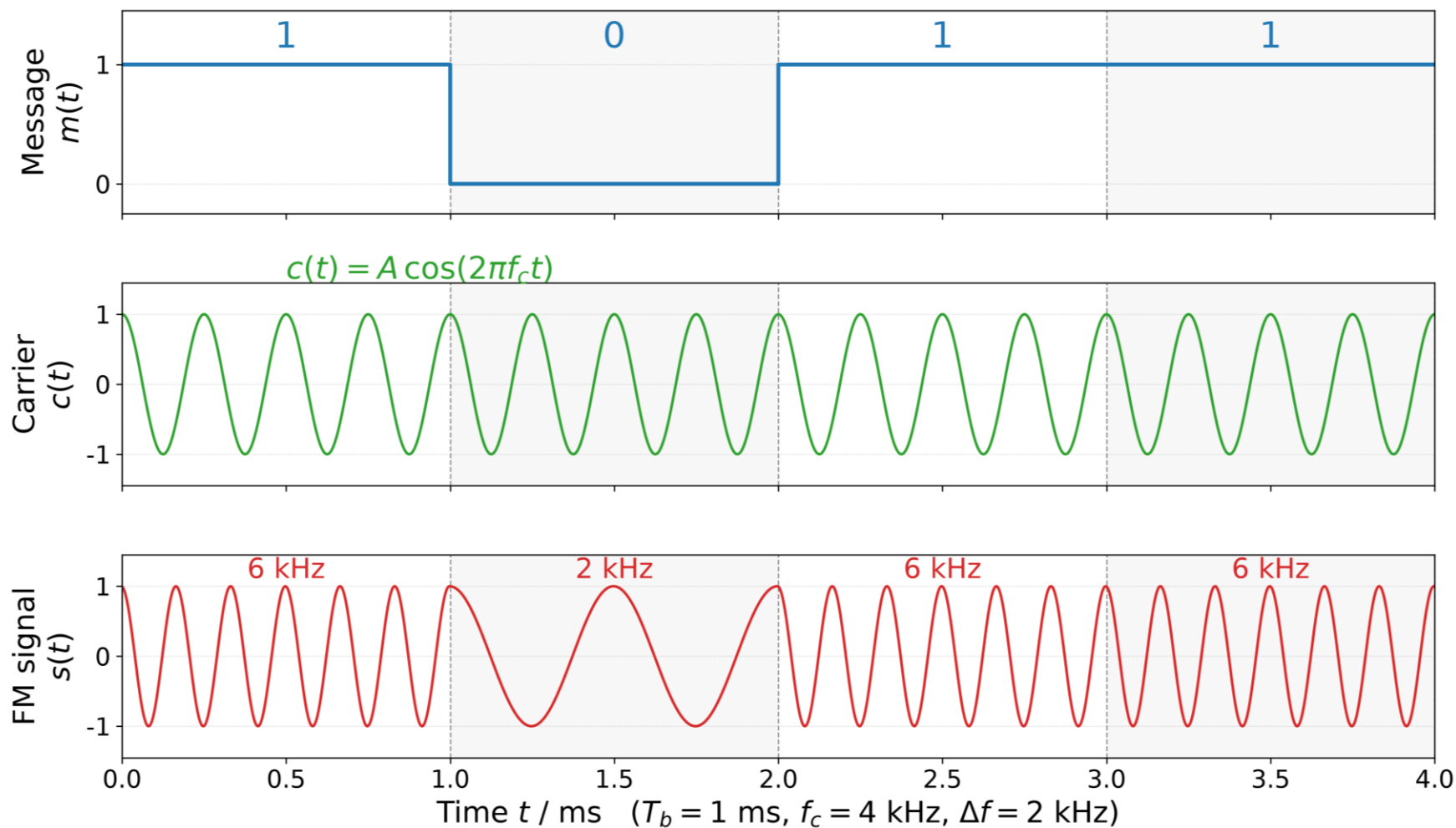
- ❑ Change the amplitude of the carrier according to the information signal.



ASK: $s(t) = A m(t) \cos(2\pi f_c t)$, $m(t) \in \{0, 1\}$ (carrier ON for bit 1, OFF for bit 0)

Frequency Modulation

❑ Change the frequency of the carrier according to the information signal.



2. Internet: IP and Routing

Scenario: Multiple Users Communicate via Internet

- ❑ Any two users/devices may exchange information.
- ❑ How can the message be delivered reliably to their destinations?
- ❑ The scenario is similar to the traditional postal system.

Internet: Network of Networks

❑Originally named ARPANET.

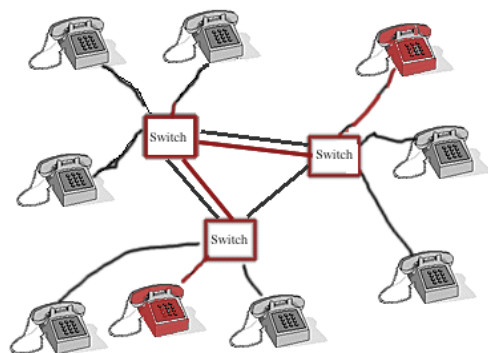
- Built in the late 1960s by ARPA (Advanced Research Projects Agency), an agency of the U.S. Department of Defense.

❑Goal: connect research computers at universities and labs.

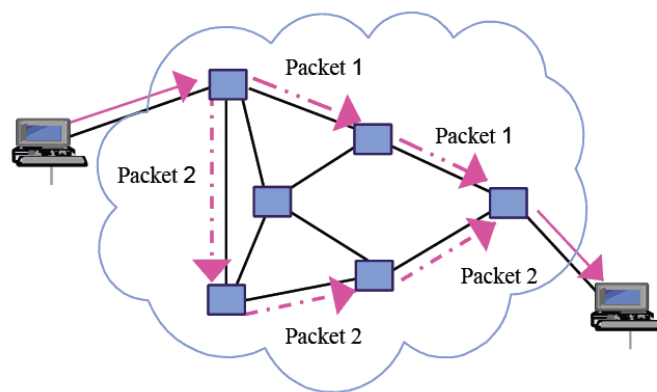
- First message sent October 29, 1969 (between UCLA & Stanford).

❑ARPANET proved packet-switched networking works.

- Circuit switching: Dedicated path is established between sender and receiver.
- Packet switching: Data is divided into packets.

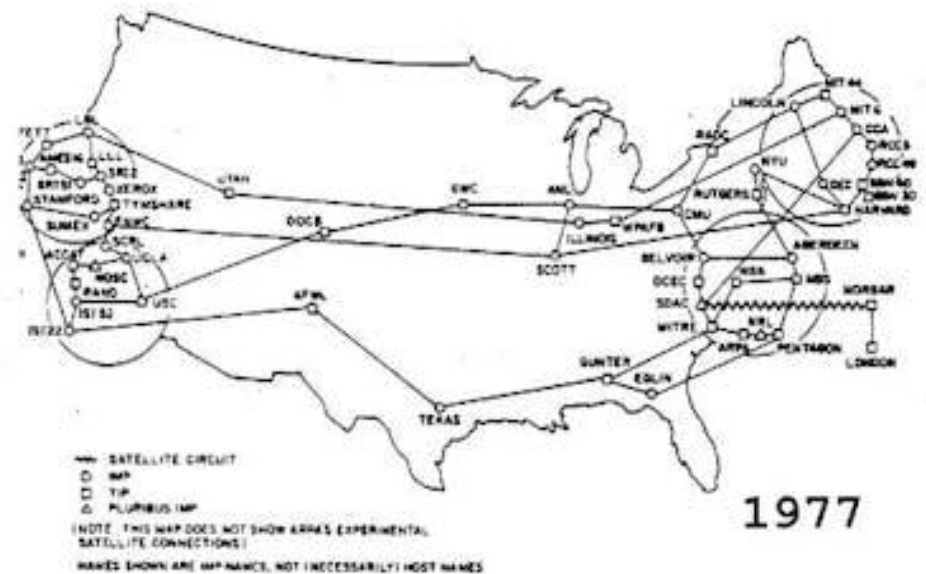
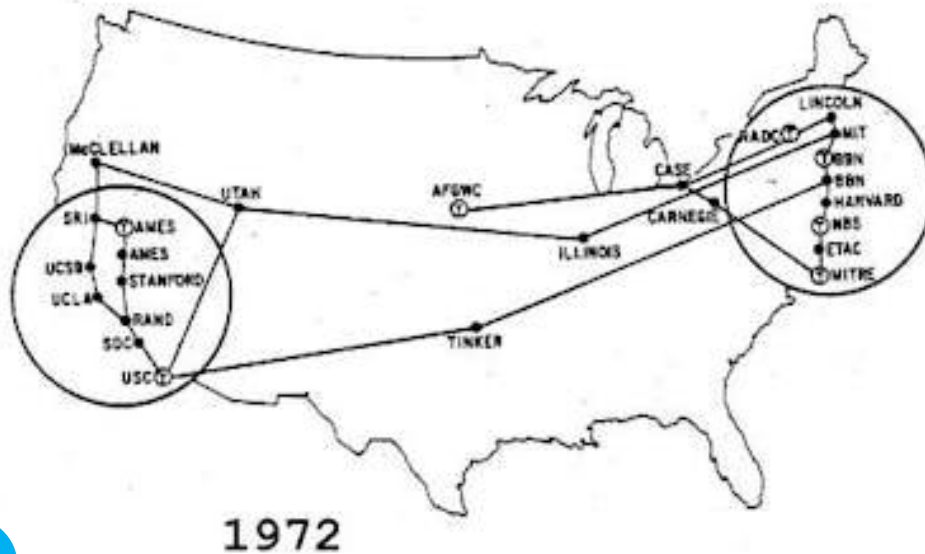
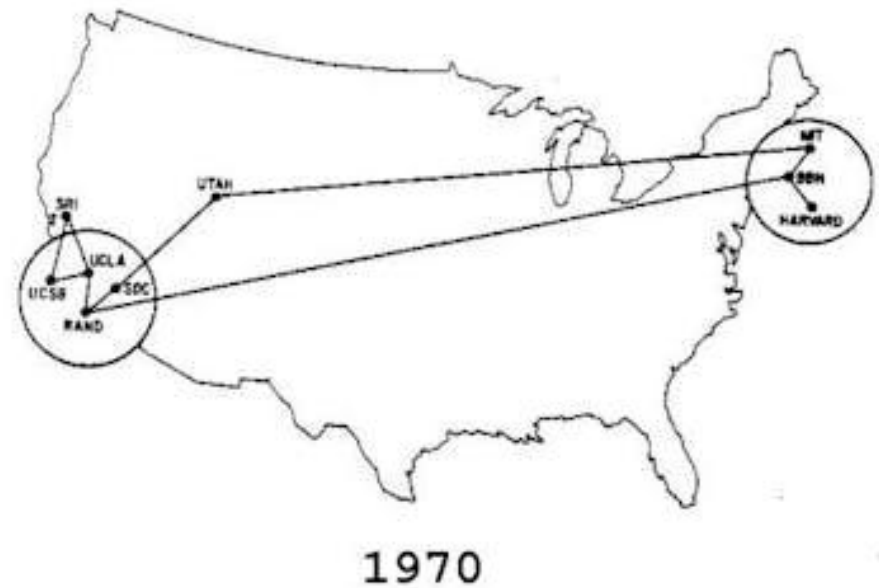
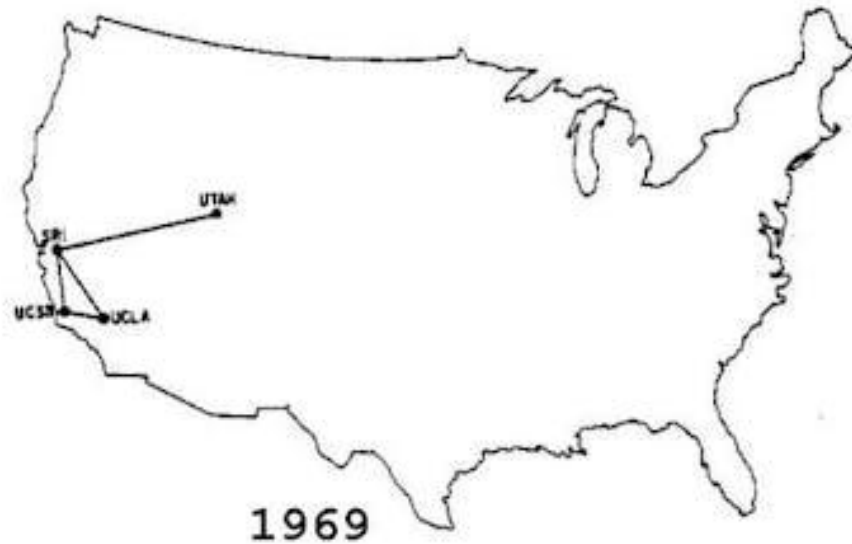


circuit switching



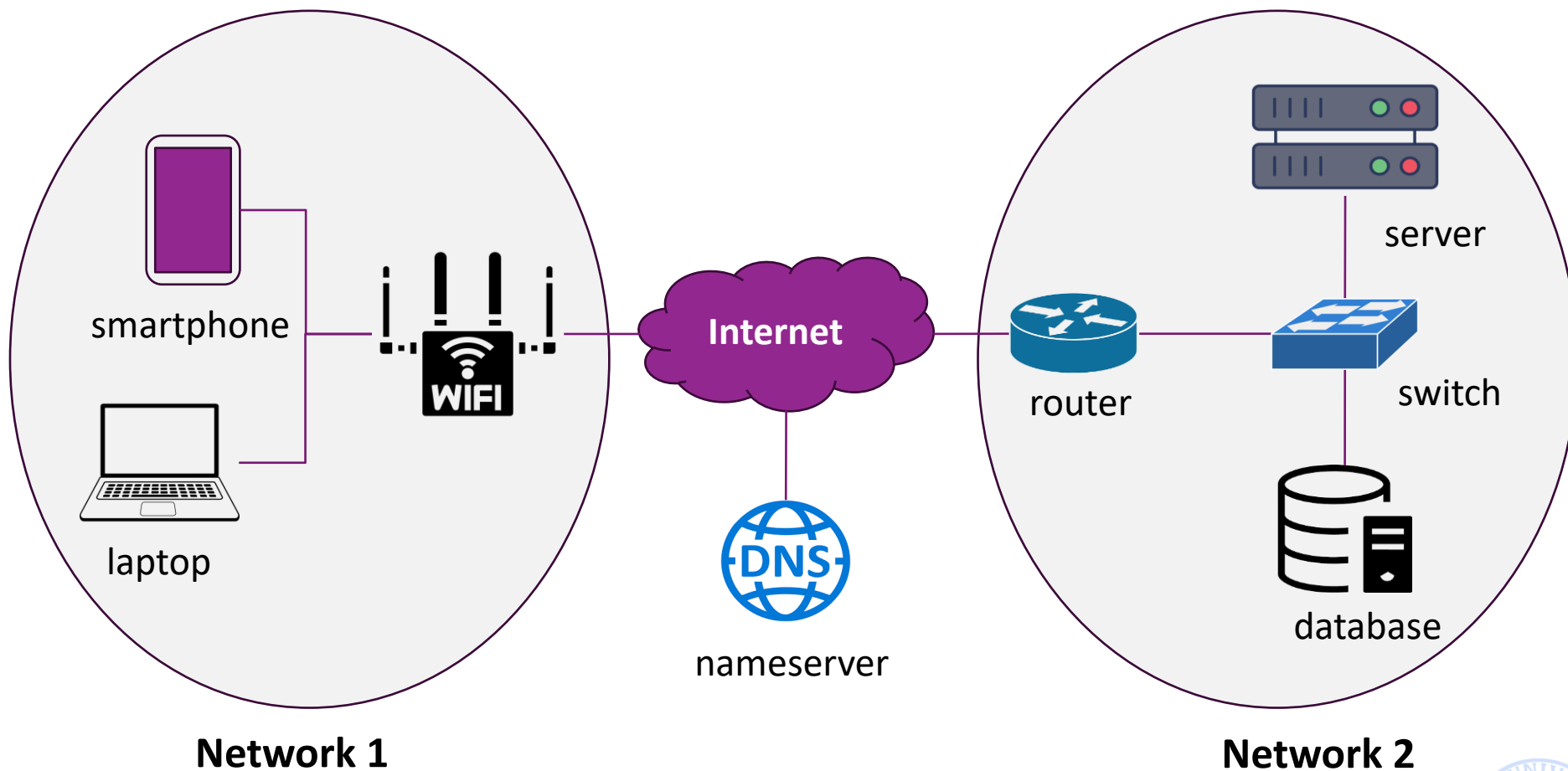
packet switching

ARPANET's Rapid Growth After Establishment



Problem Scenario

❑ Communication between a smartphone/laptop and a server.



Detailed Questions and Techniques

❑Q1. How is a device identified on a network?

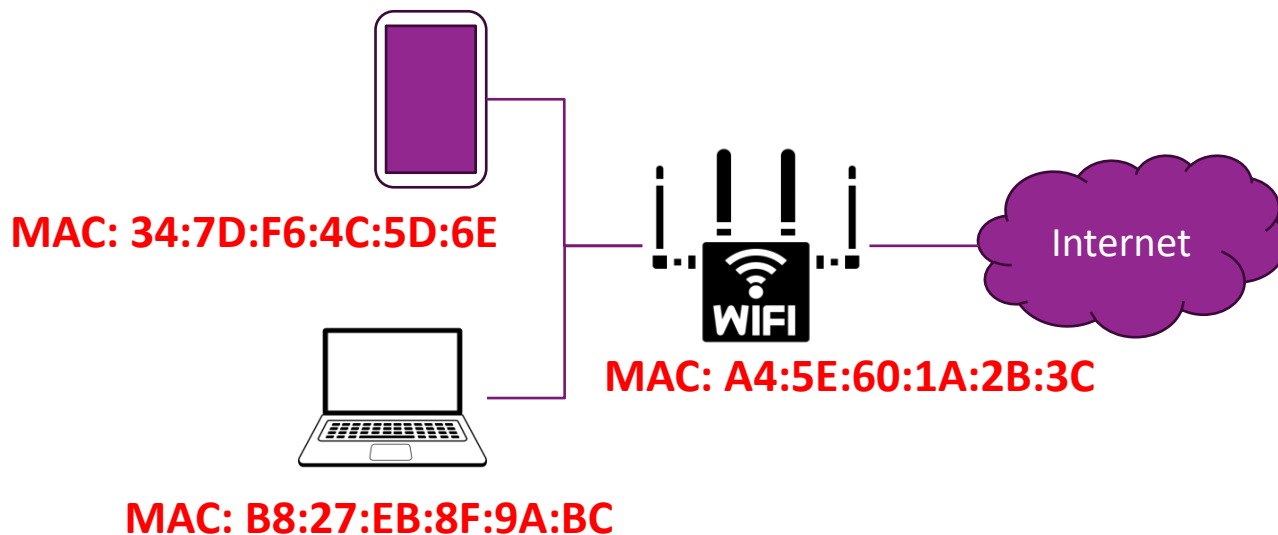
- **MAC address:** A unique identifier associated with a network interface.
 - Similar to the receiver personal ID.
- **IP address:** A logical address that identifies a device's location in the network.
 - A device can move between different networks.
 - Similar to the physical address or postal code.
- **A device with a unique MAC address can be configured with different IP addresses.**
- **Address Resolution Protocol:** To map the MAC addresses with IP addresses.

❑Q2. How does data reach a destination in another network?

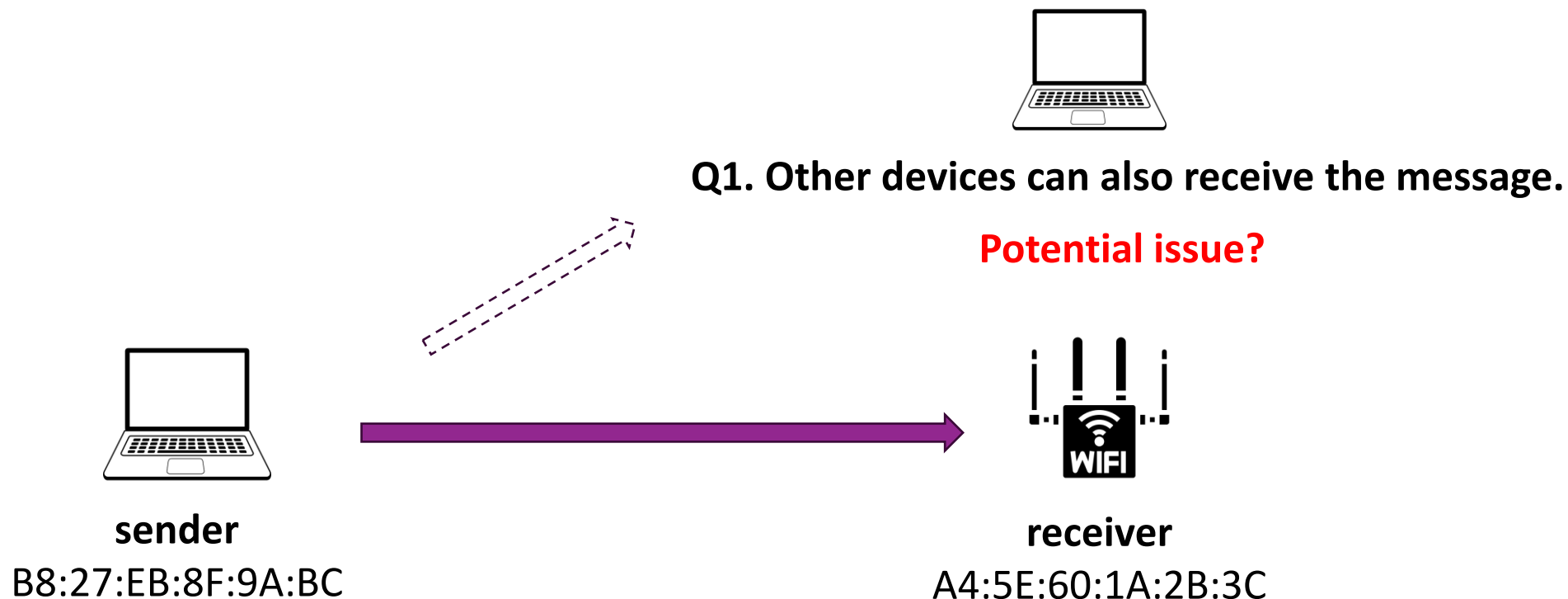
- **Routing** among different networks via the routing table.

MAC (Media Access Control) Address

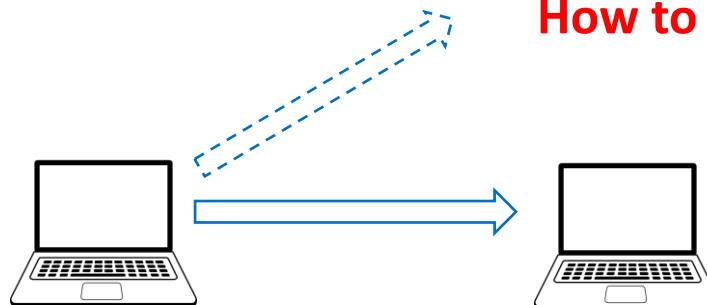
- ❑ Each network interface controller (NIC) has a unique MAC address.
- ❑ A MAC address has 48 bits (Similar to citizen ID).
 - First 24 bits: assigned to the manufacturer or organization.
 - Last 24 bits: assigned by the manufacturer or organization.
- ❑ Used for communication within a local network.



Discussion



Q2. The receiver may falsely receive other messages.
How to handle?



IP (Internet Protocol) Address

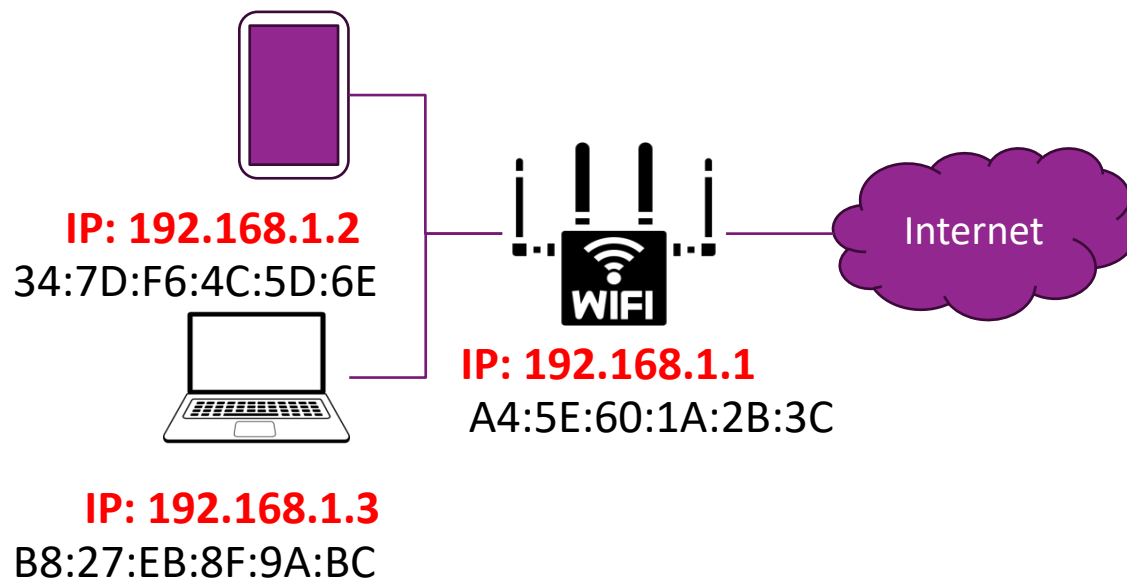
❑ Each device can be configured with different IP addresses in different networks.

❑ IPv4 (Version 4) address: *e.g.*, 192.168.1.1

➤ 32 bits long, and it is usually written as four decimal numbers separated by dots.

❑ IPv6 (Version 6) address

➤ 128 bits long and can support a vastly larger number of addresses.



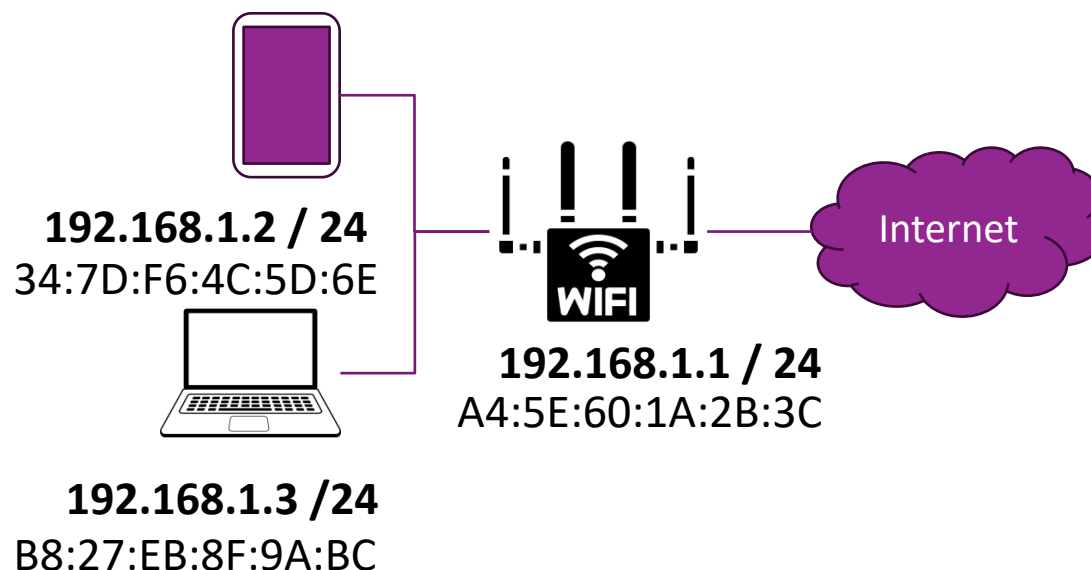
Local Area Network (LAN)

❑ MAC addresses are used for communication within a LAN.

❑ How to determine whether two devices belong to the same LAN?

➤ Physically, they are connected to the same local network.

➤ Logically, their IP addresses must share the same prefix or network portion.



IP Addresses and Networks

❑ An IP address has two parts.

- **Network prefix:** The left n bits that identify the network.
- **Host part:** The rest bits identifies the device on that network.

❑ How to determine the network part?

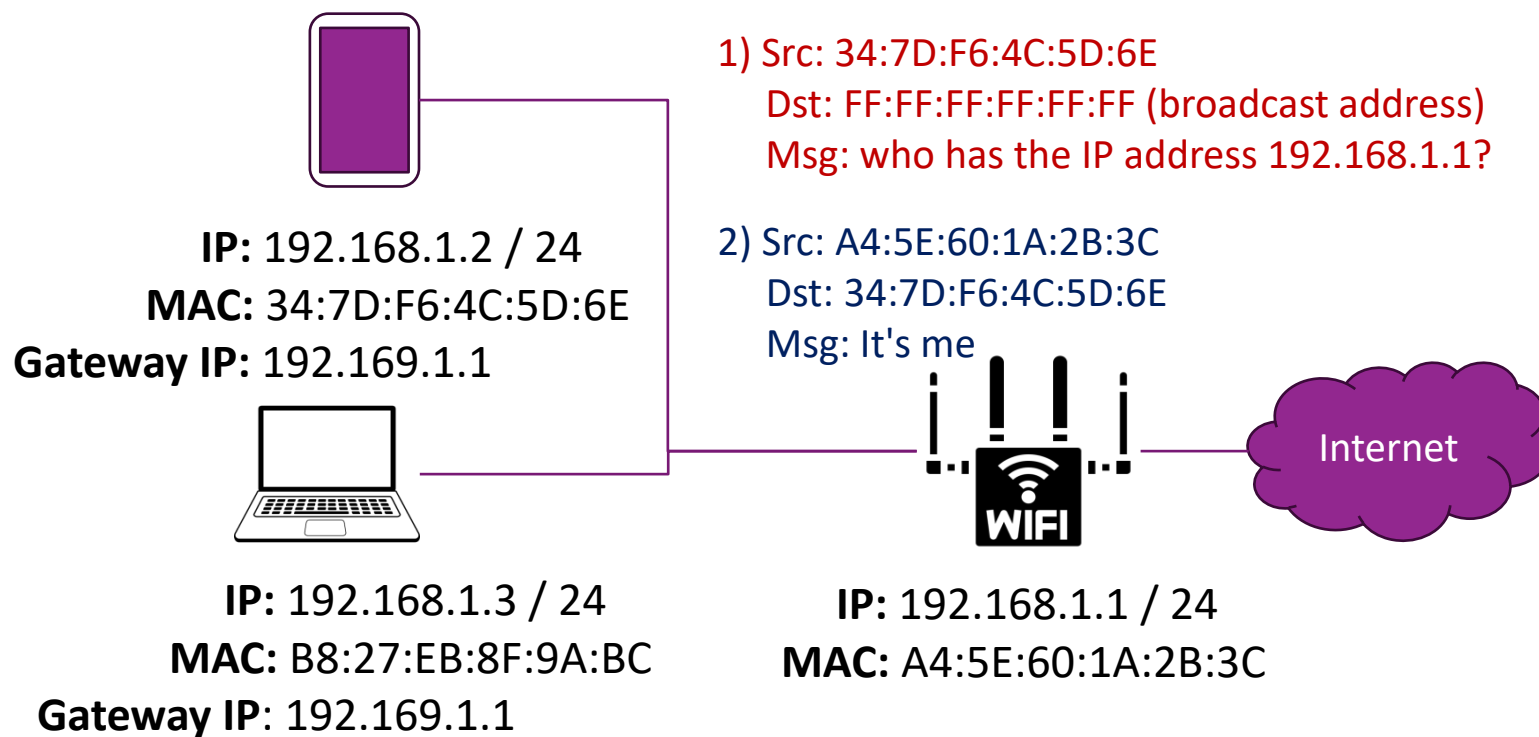
- CIDR notation, *e.g.*, 192.168.1.1 / 24 (number of prefix bits).
- Uses subnet masks, *e.g.*, 192.168.1.1 / 255.255.255.0.
 - Mask bit = 1: this bit must match (the address bits must be identical).
 - Mask bit = 0: this bit can vary (no need to match).

❑ IP within the same network must share the same network prefix.

- 192.168.1.1 / 24 and 192.168.1.100 / 24 are within the same network.
- 192.168.1.1 / 26 and 192.168.1.100 / 26 do not belong to the same network.
- 192.168.1.100 / 24 and 192.168.2.10 / 24 ?

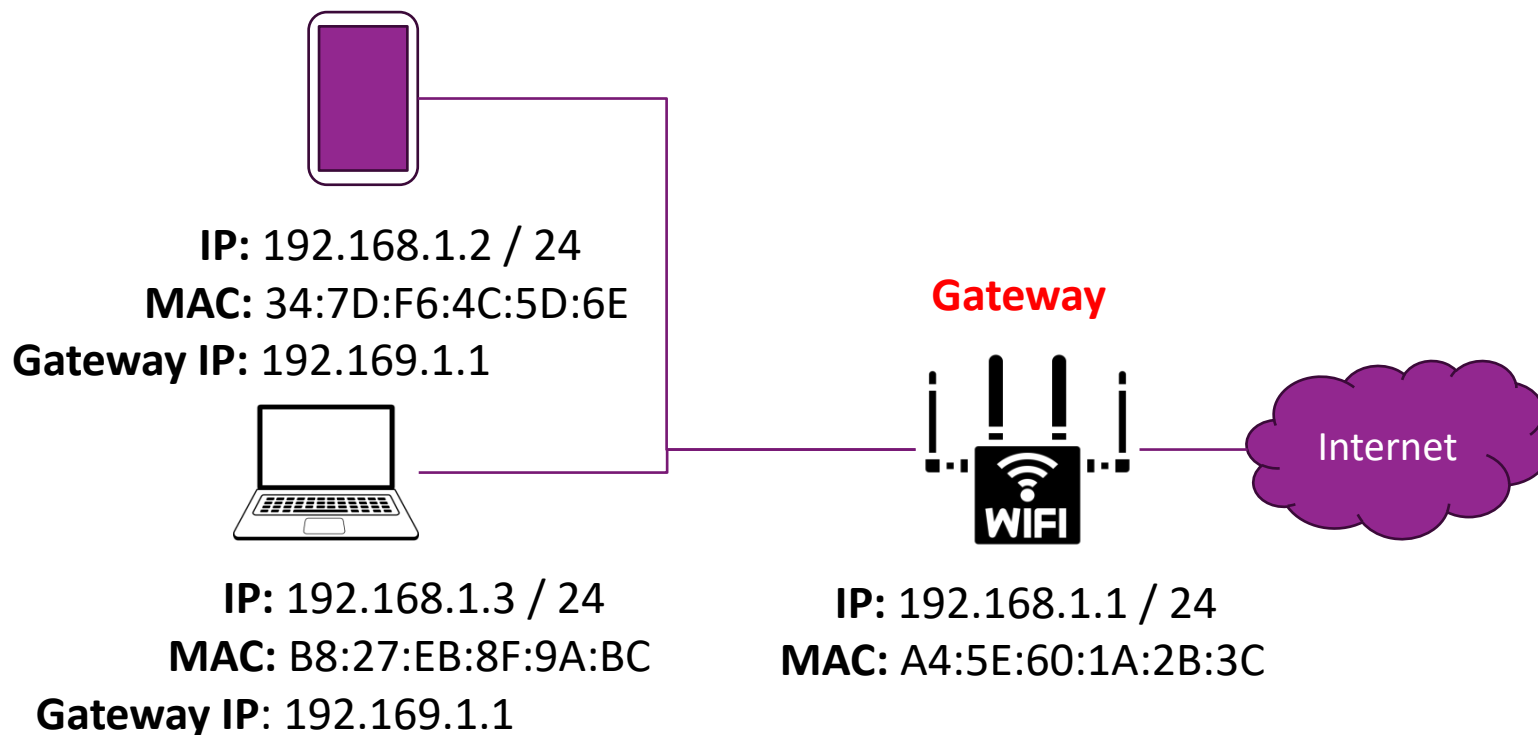
Address Resolution Protocol (ARP)

- ❑ At first, a device may know the destination's IP, but not its MAC.
- ❑ ARP resolves an IP address to the corresponding MAC address.



From LAN to the WAN (Wide Area Network)

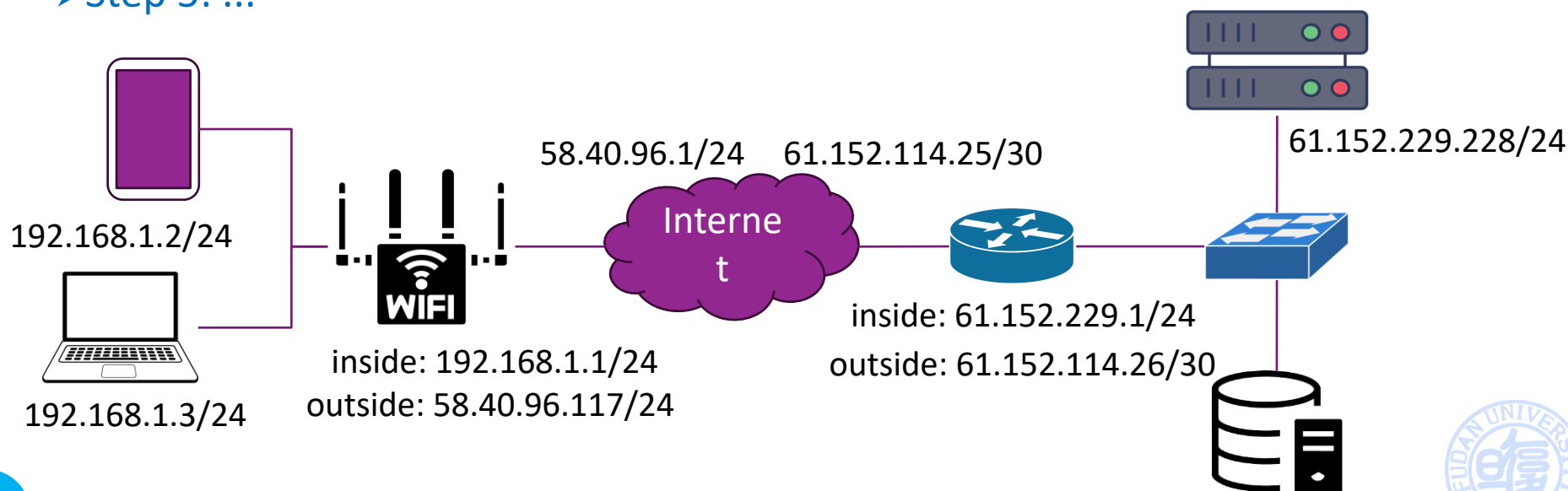
- ❑ A default gateway connects the LAN to other networks.
- ❑ Devices on the LAN are configured with the IP of the default gateway.



Routing among Different Networks

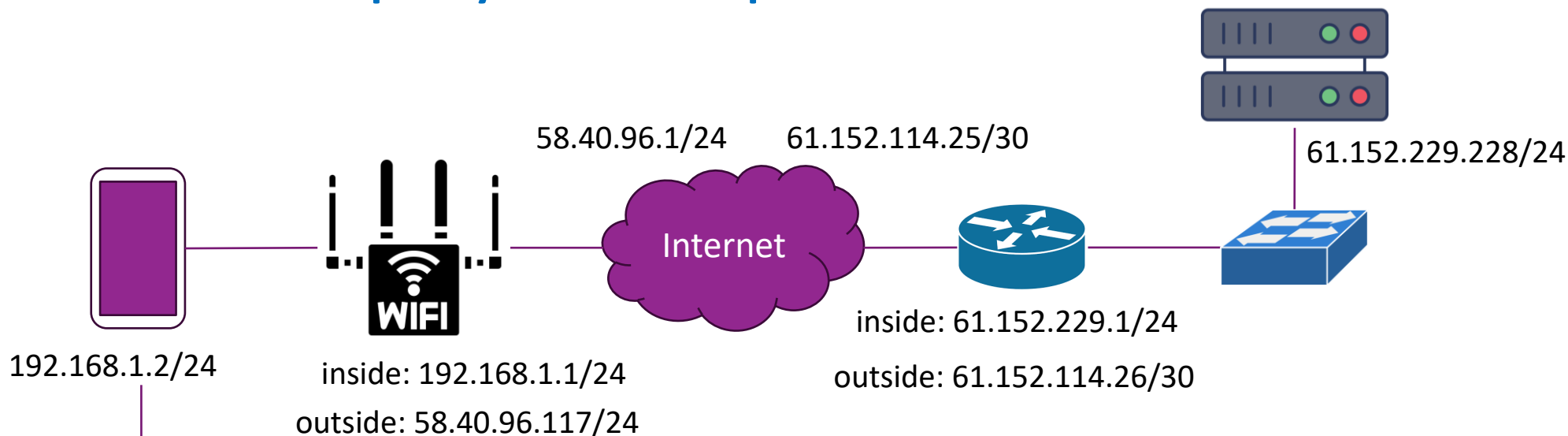
❑ Suppose from 192.168.1.2 to 61.152.229.228.

- Step 1: Client inquiries the MAC address of the gateway via ARP.
- Step 2: Client composes a message with the destination IP: 61.152.229.228.
- Step 3: Client sends the message to the gateway's MAC address.
- Step 4: The gateway checks the routing table and forwards the message via the outside port to 58.40.96.1.
- Step 5: ...



Routing Table

- ❑ Each device has a route table.
- ❑ The entries specify the next hop based on the destination.

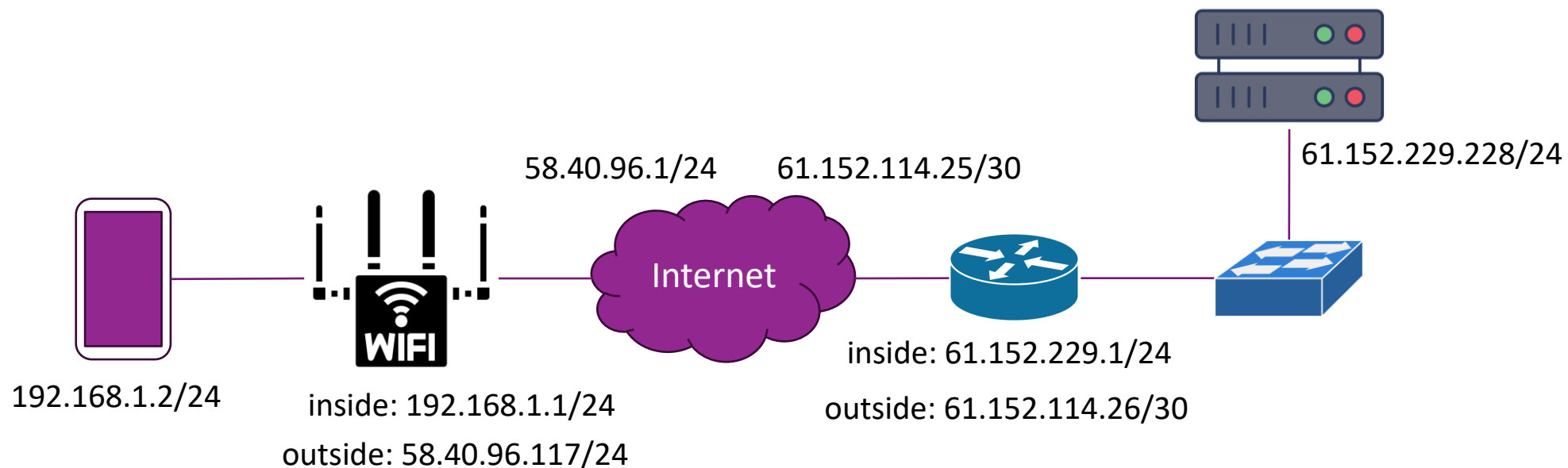


Destination	Next Hop	Interface
default	192.168.1.1	eth0

```
MacBook-Pro-2:artisan huixu$ netstat -rn  
Routing tables
```

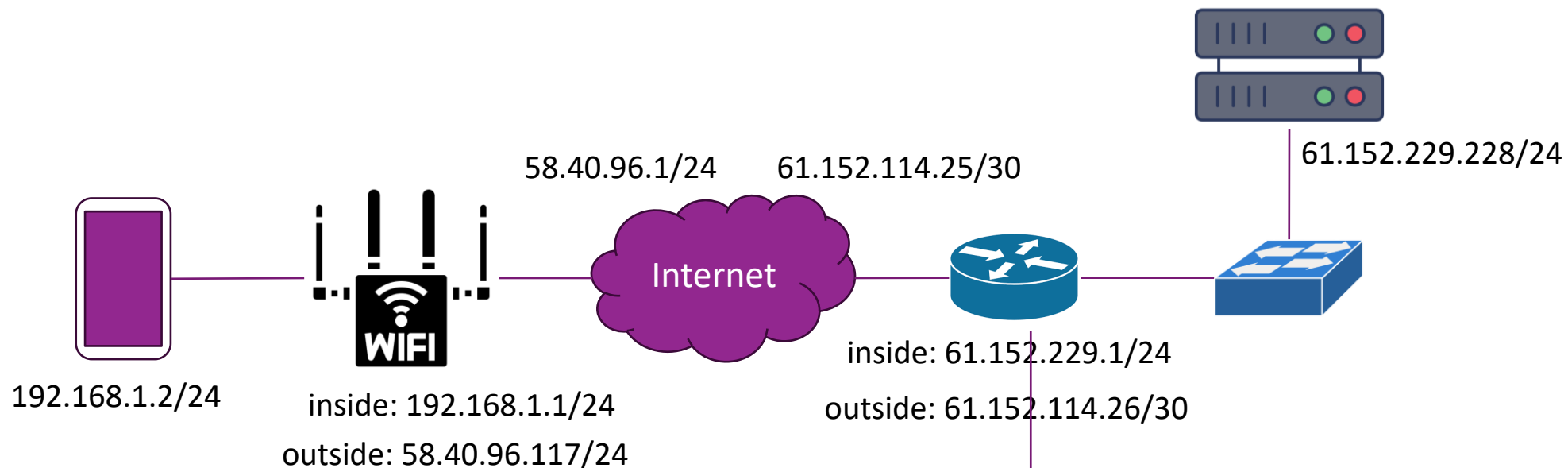
```
Internet:  
Destination      Gateway          Flags  
default          192.168.1.1     UGScg  
127              127.0.0.1       UCS  
127.0.0.1        127.0.0.1       UH
```

Routing Table



Destination	Next Hop	Interface
default	58.40.96.1	outside

Routing Table



Destination	Next Hop	Interface
61.152.229.0/24	-	inside
default	61.152.114.25	outside

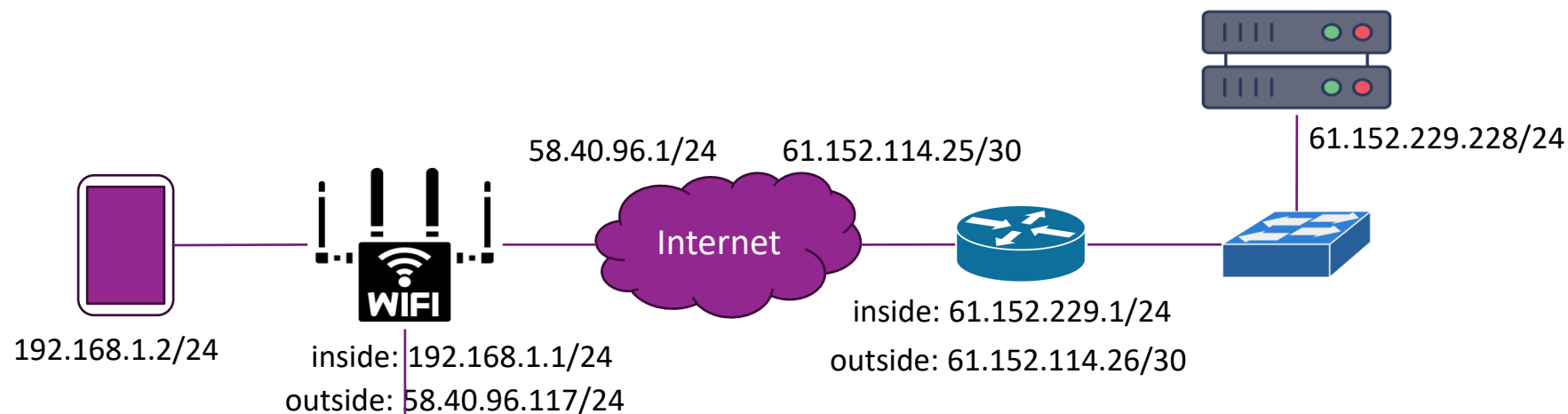
Private Network

- ❑ Three ranges of IP addresses are reserved for private network.
- ❑ Private addresses are not routable on the public Internet.
- ❑ Access Internet via a gateway with Network Address Translation.
- ❑ Widely used due to the limited number of IPv4 addresses.

Range	Next Hop	No. of Addresses
1	10.0.0.0 – 10.255.255.255	2^{24} or 16777216
2	172.16.0.0 – 172.31.255.255	2^{20} or 1048576
3	192.168.0.0 – 192.168.255.255	2^{16} or 65536

Network Address Translation (NAT)

- ❑ Private IP addresses are not routable on the public Internet.
- ❑ NAT allows devices using private IP addresses to access the public Internet.



Source Address		Translated Address	
IP	Port	IP	Port
192.168.1.2	12345	58.40.96.117	32123
192.168.1.2	12346	58.40.96.117	32124
192.168.1.2	12347	58.40.96.117	32125

Internet

- ❑ **Composed of many autonomous systems (AS).**
- ❑ **Each AS is an independent network entity, such as**

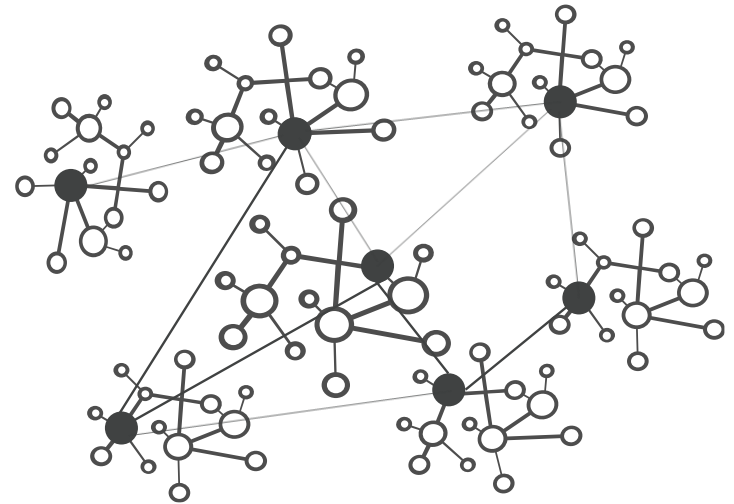
- Internet Service Provider (ISP)
- Large high-tech organizations

- ❑ **Within an Autonomous System:**

- Multiple routers manage multiple networks
- Uses OSPF for internal routing

- ❑ **Interconnection between Autonomous Systems:**

- Use BGP for communication between different ASes



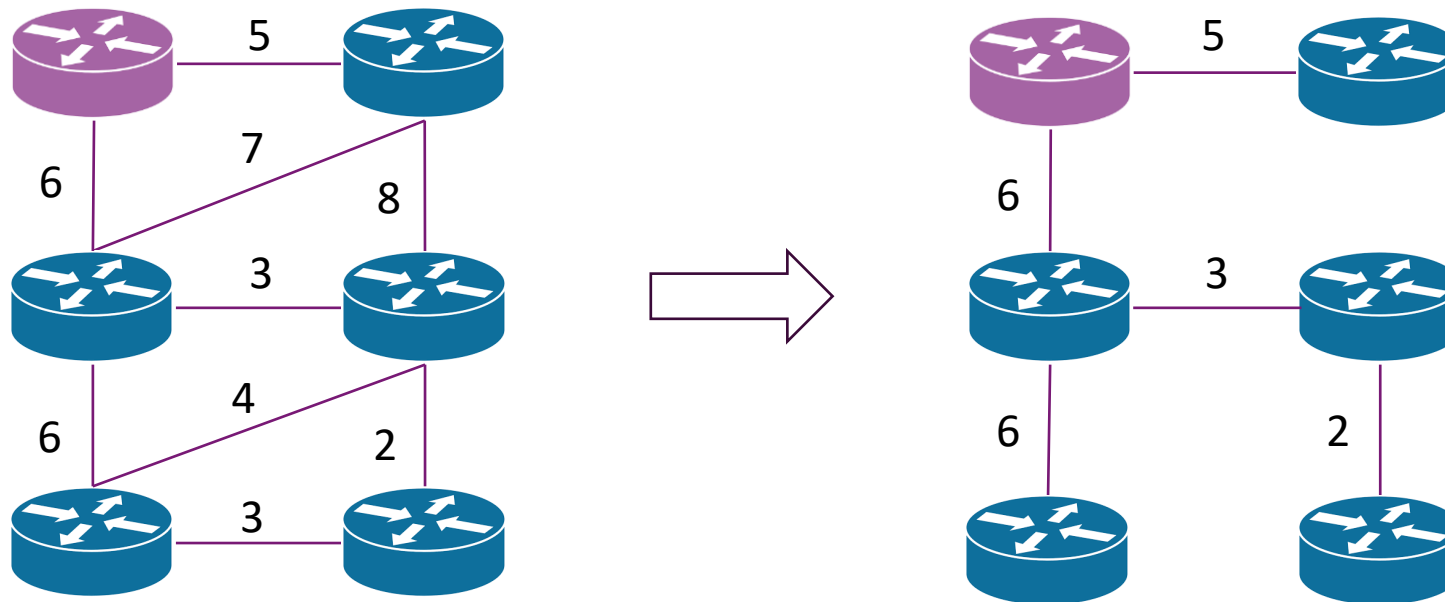
Open Shortest Path First (OSPF)

❑ OSPF is a dynamic routing protocol.

❑ Build the routing table within the same autonomous systems.

➤ Find the shortest path to each destination, *e.g.*, based on Dijkstra's algorithm.

➤ Convert a network topology to a spanning tree.



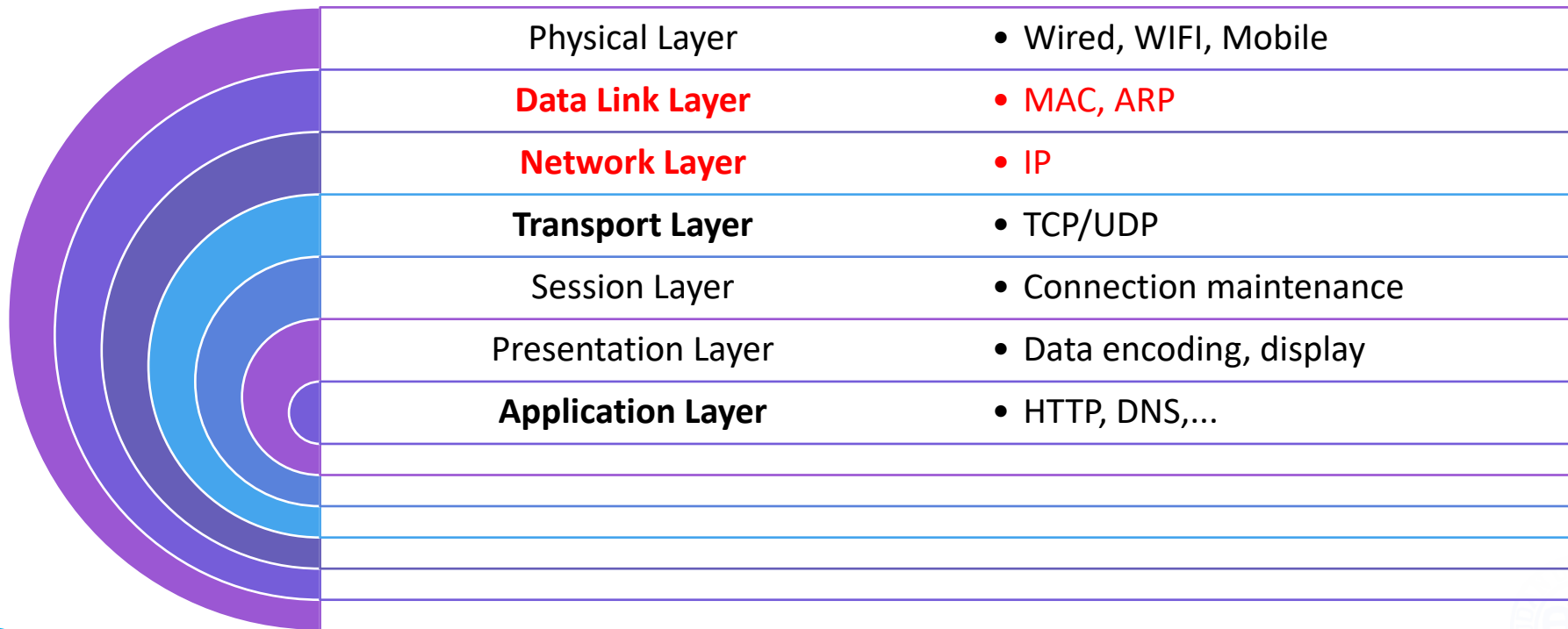
Example AS

Rank ↑	AS Number	Organisation	Total IPv4 Addresses
1	AS749	US United States Department of Defense (DoD)	222,267,882
2	AS16509	US Amazon.com, Inc.	157,616,898
3	AS4134	CN No.31,Jin-rong Street Beijing 100032	98,033,046
4	AS7018	US AT&T Enterprises, LLC	89,909,732
5	AS8075	US Microsoft Corporation	80,698,305
6	AS4837	CN CHINA UNICOM China169 Backbone	57,217,679
7	AS4766	KR Korea Telecom	46,335,386
8	AS7922	US Comcast Cable Communications, LLC	42,055,680
9	AS701	US Verizon Business	40,515,150
10	AS17676	JP SoftBank Mobile Corp. Tokyo Port City Takeshiba ..	38,309,290

3. Messaging

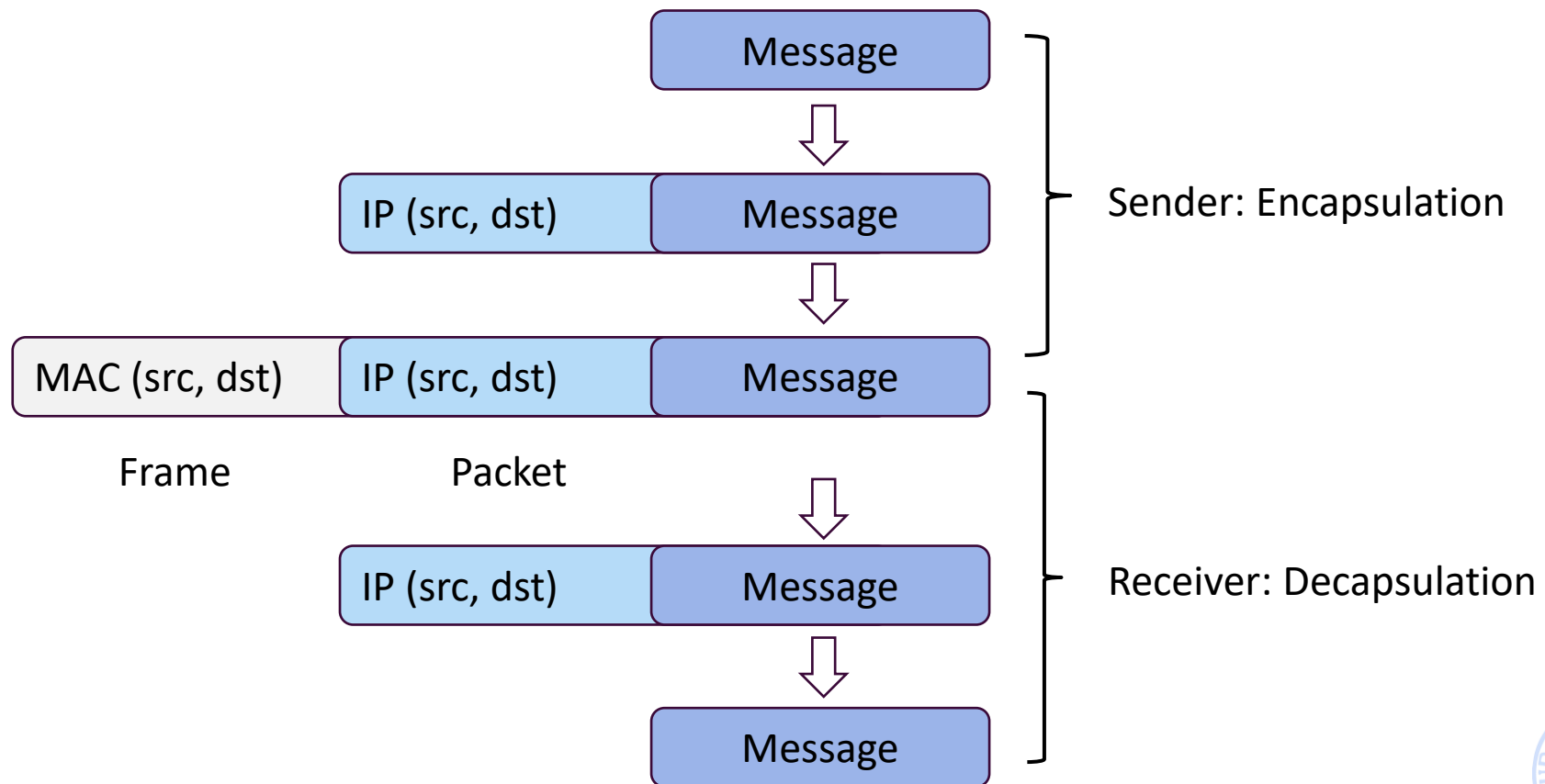
The OSI Model

- ❑ A 7-layer framework for understanding network communication.
- ❑ Developed by ISO to standardize networking.
- ❑ Each layer is decoupled from the others, allowing independent development and troubleshooting.



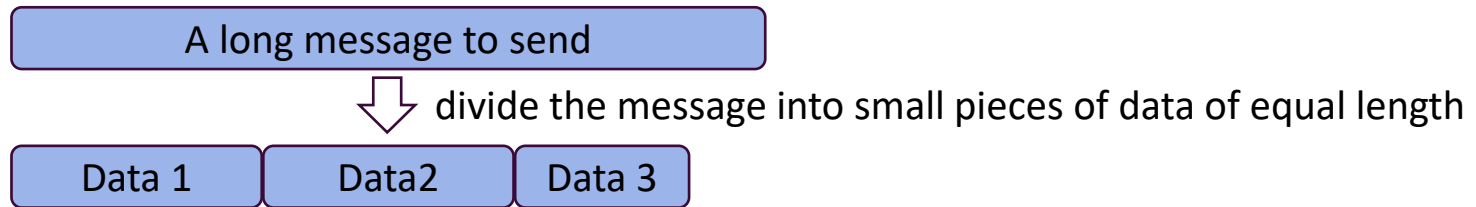
Message Encapsulation/Decapsulation

- ❑ **Encapsulation:** Each layer adds its own header to the data it receives from the layer above.
- ❑ **Decapsulation:** Each layer removes its own header.

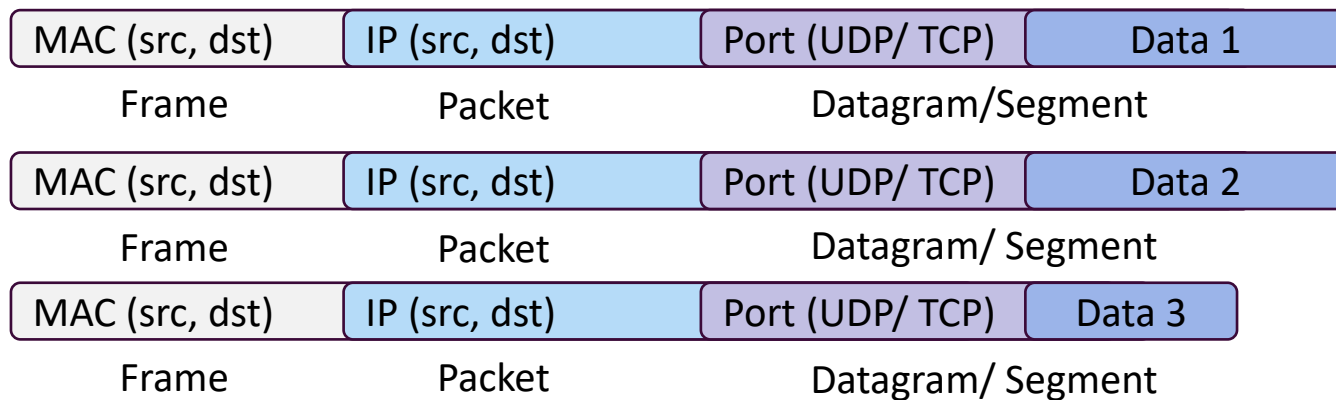


Fragmentation

- ❑ A network cannot handle packets larger than a certain size (MTU – Maximum Transmission Unit).
- ❑ If a message is too long, it is divided into smaller, equal-sized pieces called fragments.



Encapsulate each piece of data:



Transmission Control Protocol (TCP)

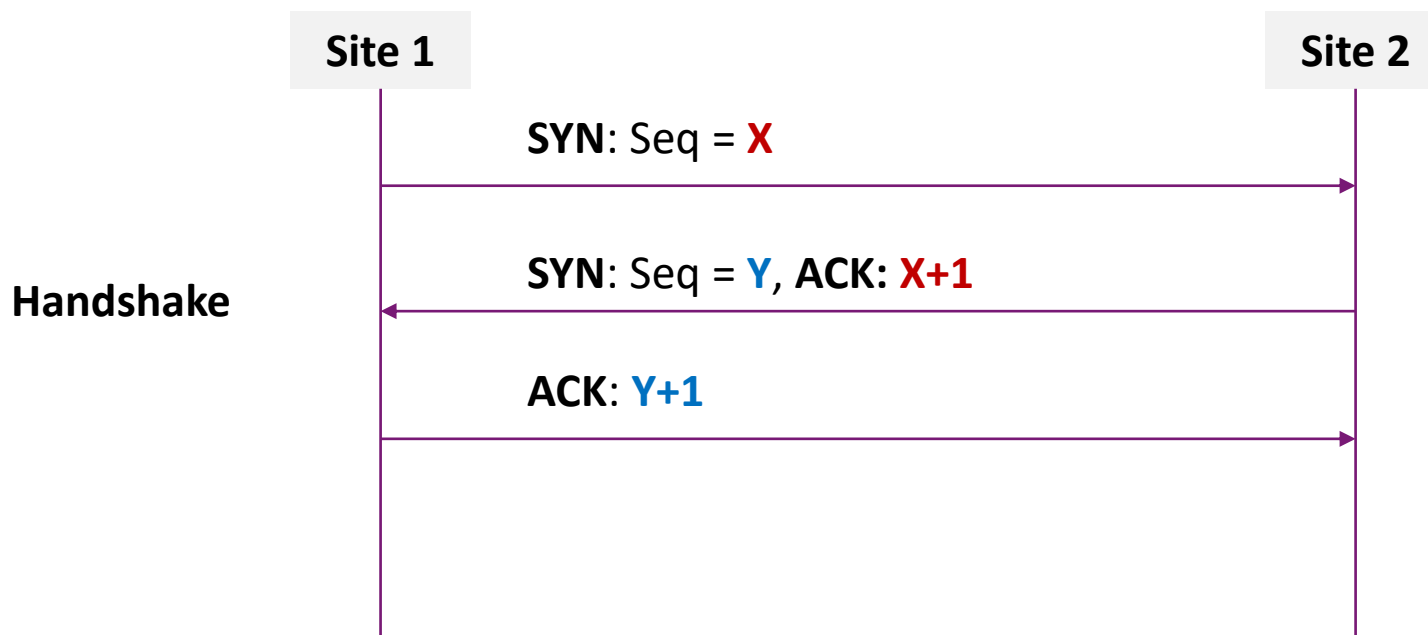
- ❑ Data (segments) can be lost during transmission over the network.
- ❑ TCP provides reliable, ordered, error-checked delivery of data.
- ❑ Turn the Internet's "best effort" IP service into a virtual connection.
- ❑ Three phases:
 - Handshake: Establish the connection between sender and receiver
 - Data Transfer: Send and receive data reliably
 - Termination: Close the connection gracefully

TCP: Three-Way Handshake

❑ Establish a reliable connection before sending data.

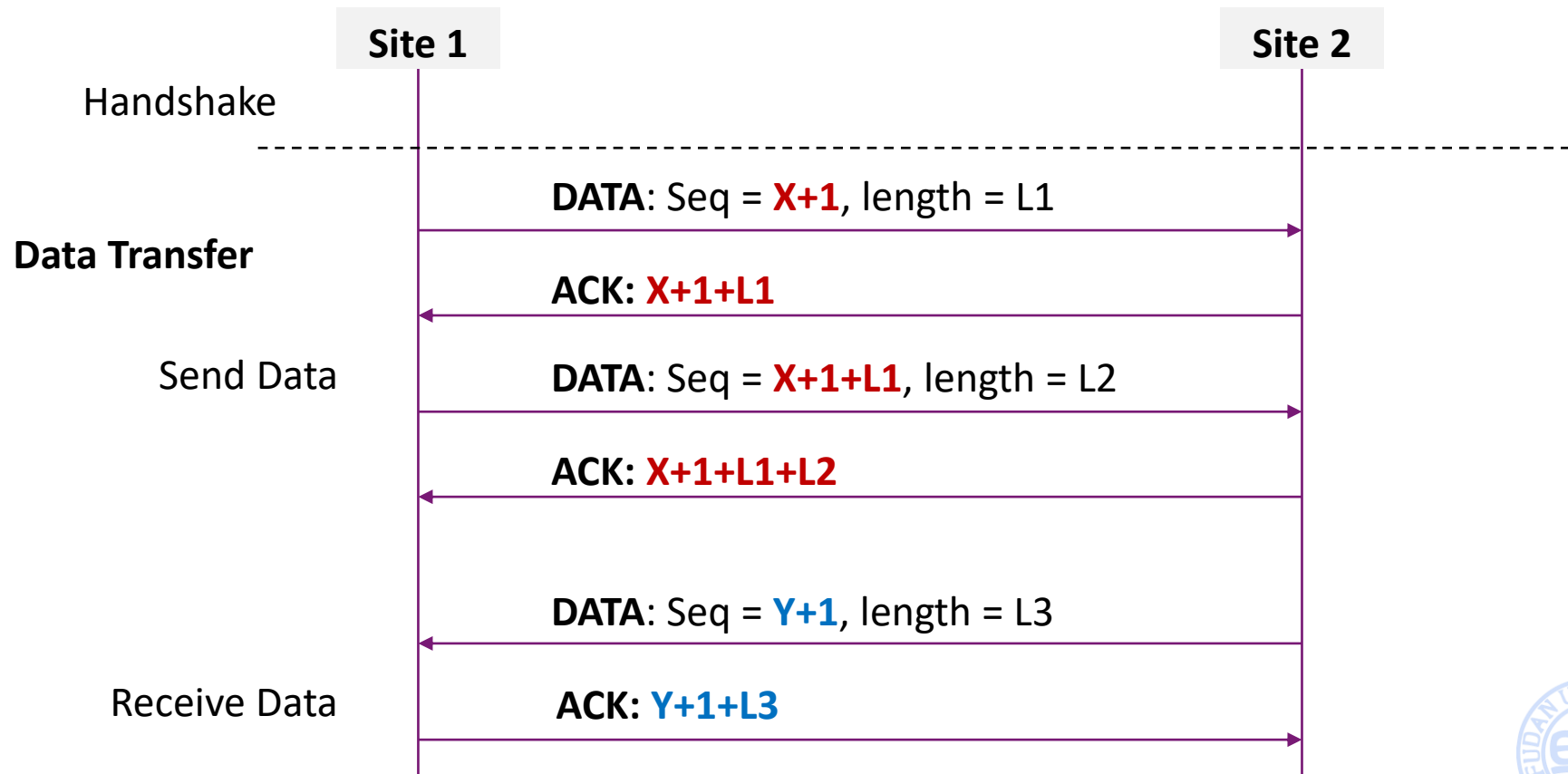
- Site 1 sends a packet with the SYN flag set and a random sequence number.
- Site 2 responds with SYN + ACK, acknowledging the client's sequence number and providing its own sequence number.
- Site 1 sends an ACK, confirming the server's sequence number.

❑ ACK also indicates the next sequence number it expects.



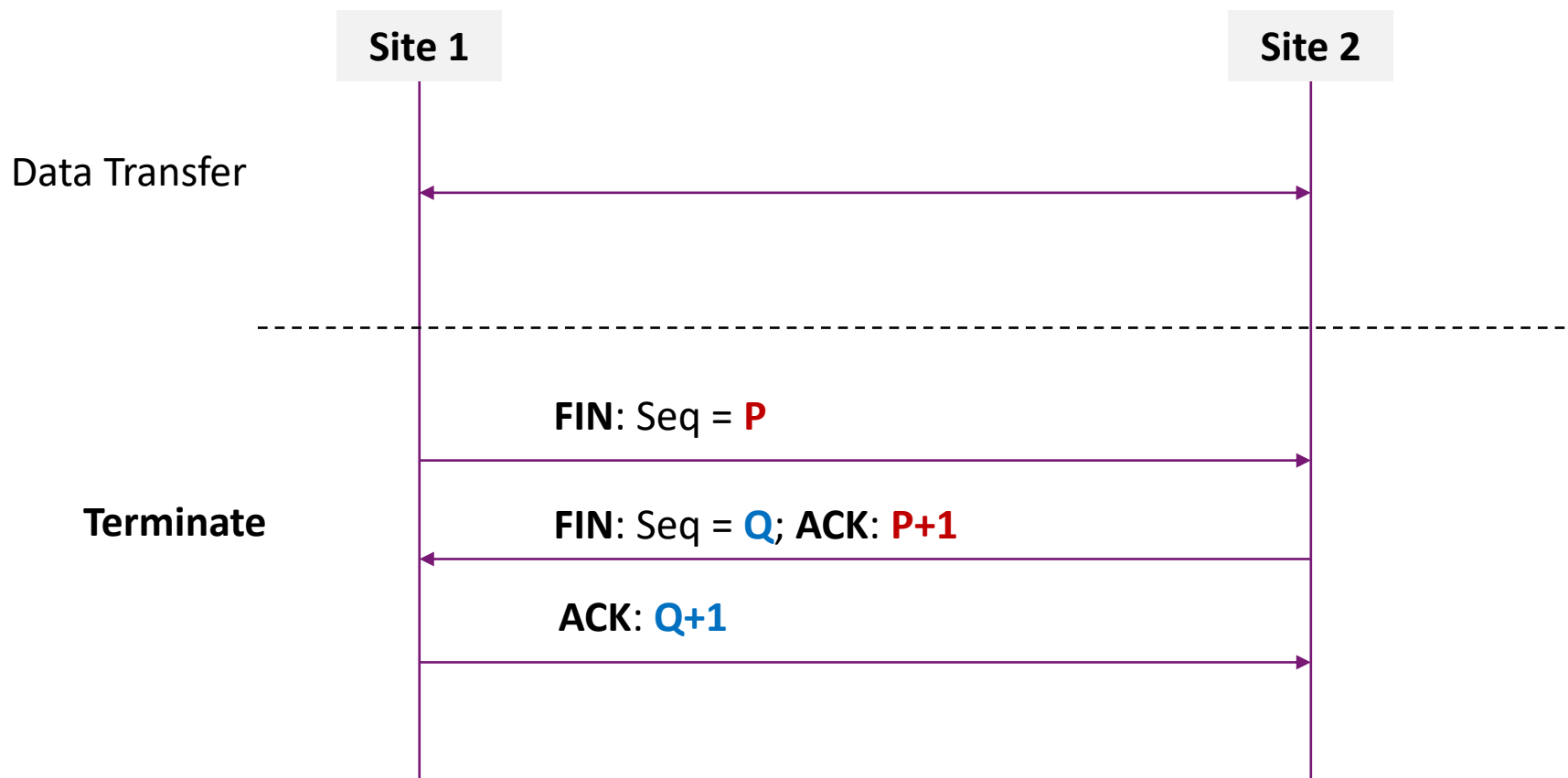
TCP: Data Transfer

- ❑ Sender sends segment with $\text{Seq} = N$, containing bytes N to $N+L$.
- ❑ Receiver receives it and sends $\text{ACK} = X+L+1$, indicating it successfully received those bytes.



TCP: Close

❑ The sender must ensure that all segments are received correctly.



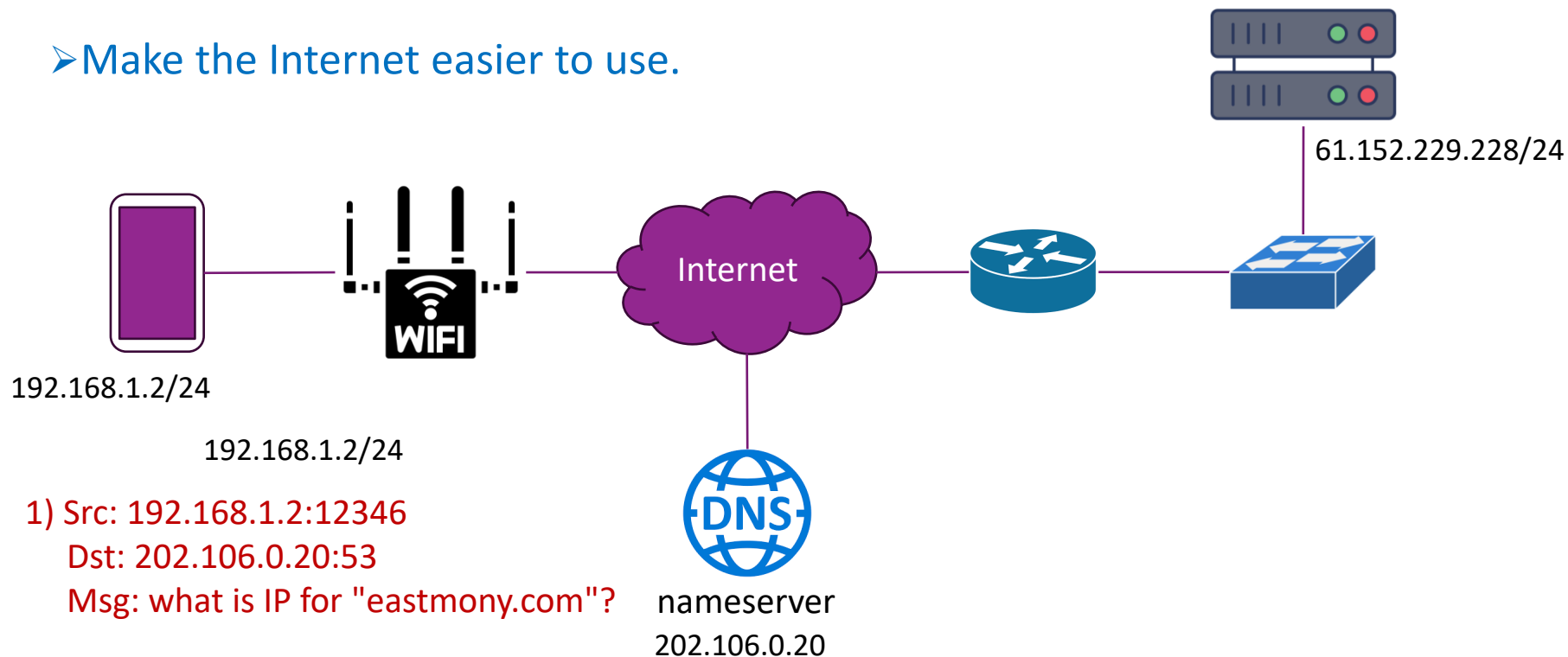
Domain Name Service (DNS)

❑ IP address is still hard to remember.

❑ DNS acts like a phone book for the Internet.

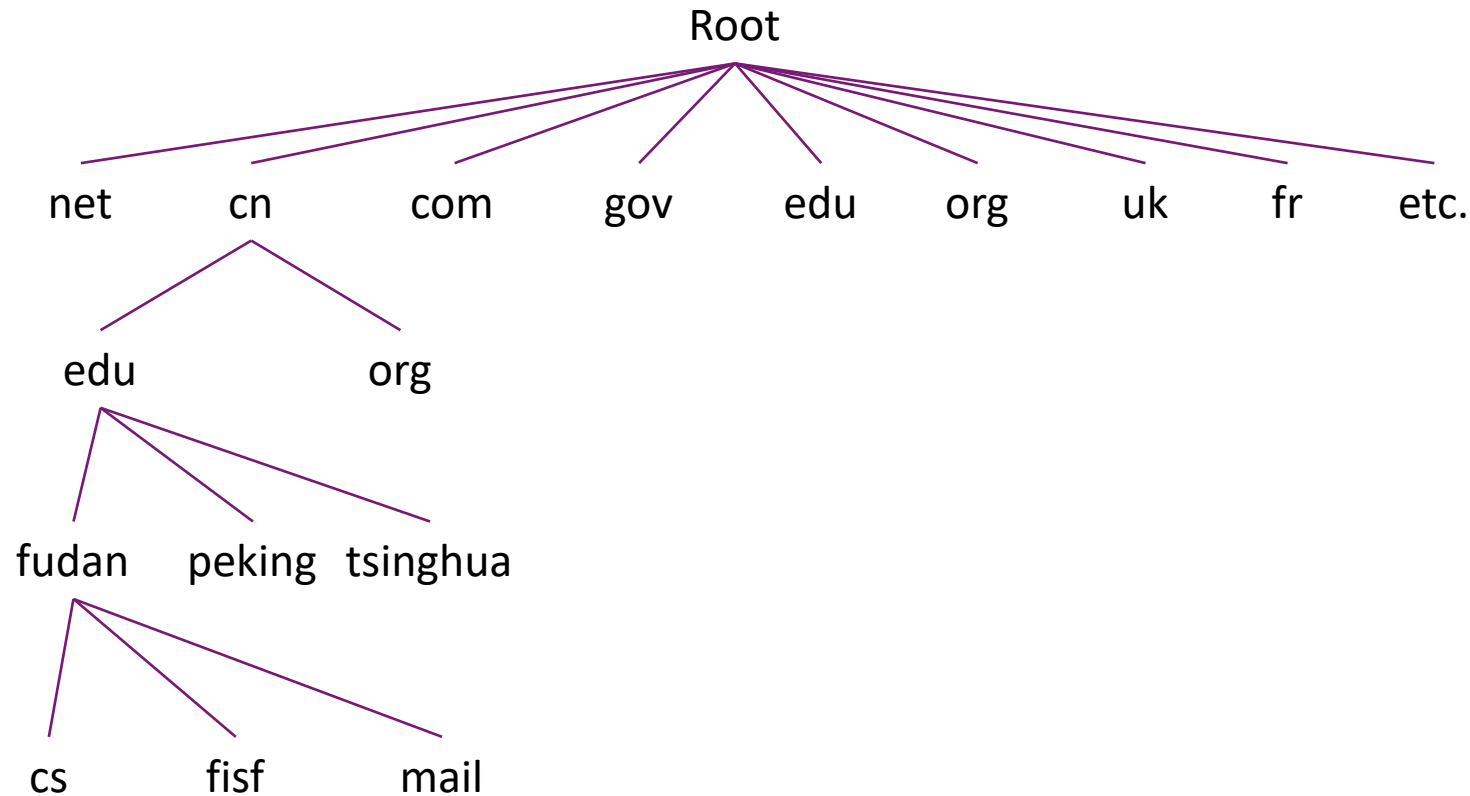
➤ Translate human-readable domain names into IP addresses.

➤ Make the Internet easier to use.



Domain Name in a Tree Structure

- ❑ Domain names are organized in a tree-like hierarchy.
- ❑ Each node is managed by an organization.



Root DNS Servers

□ 13 root servers (A-M) and hundreds of mirrors around the world.



Leaflet | Map data © OpenStreetMap contributors

Hypertext Transfer Protocol (HTTP)

❑ **An application-layer protocol used by web browsers and servers.**

- Work on top of TCP/IP.
- Transfer web pages, images, and other resources over the Internet.

❑ **Define how browsers request and servers respond with content.**

➤ Request (from client):

- URL: the address of the resource
- Method: GET, POST, PUT, DELETE, etc.
- Headers: additional info (e.g., browser type, cookies)

➤ Response (from server):

- Status code: 200 OK, 404 Not Found, 500 Internal Server Error
- Headers: content type, length, cookies, etc.
- Body: the actual content (HTML, JSON, image...)

HTTP: An Example

1) Client (browser) sends a request.

```
GET /about HTTP/1.1  
Host: www.example.com  
User-Agent: Firefox/141.0
```

2) Server processes it and sends back a response.

```
HTTP/1.1 200 OK  
Content-Type: text/html  
<html>  
  <h1>Hello, world!</h1>  
</html>
```

Practice with Agents

❑ Trace the network path from your computer to a target network.

- Identify the IP address of each hop along the path.
- Identify the ISP or organization associated with each IP address.

Network path to amazon.com with ISP per hop

Shell: Look up ISP/org for each public hop IP

Here's the full path to **amazon.com (98.87.170.74)**, traced over 30 hops (many later hops dropped ICMP, so the trace ends at hop 20 but the destination resolved in DNS):

Hop	IP	Latency	ISP / Organization	Location
1	10.110.248.83	~33 ms	Private (RFC1918) — local router	LAN
2	2.255.255.253	~34 ms	Arelion (Telia Company) — AS1299	Solna, SE (edge)
3	10.110.248.81	~45 ms	Private (RFC1918)	LAN
4	10.250.3.177	~38 ms	Private (RFC1918)	LAN
5	10.250.2.106	~36 ms	Private (RFC1918)	LAN
6	202.120.255.249	~36 ms	CERNET — AS4538 (Shanghai Jiao Tong Univ.)	Shanghai, CN
7	101.4.116.61	~43 ms	CERNET — AS4538	Beijing, CN

4. More Practice: Packet Analysis

Practice: Network Traffic Analysis with Wireshark

❑ Download and install Wireshark.

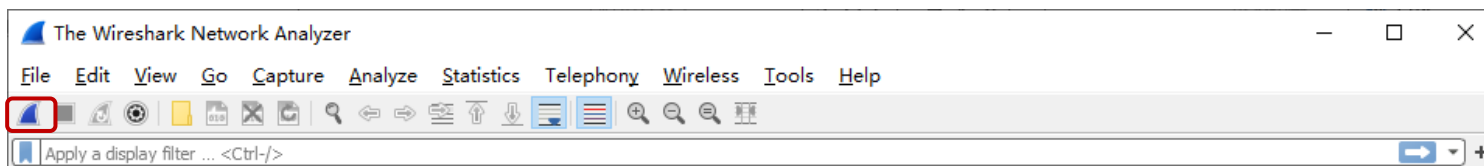
➤ <https://www.wireshark.org/download.html>

❑ Access a website (e.g., www.fudan.edu.cn) and capture the network traffic with Wireshark.

➤ Filter related packets.

➤ Interpret the meaning of each related packets.

Start Packet Capturing with Wireshark



2. Click the button to start

Open

C:\Users\china\Documents\wechat.pcapng (21 KB)

Capture

...using this filter:

All interfaces shown

Local Area Connection* 10	_____
Local Area Connection* 7	_____
Local Area Connection* 1	_____
Bluetooth Network Connection	_____
Wi-Fi	~~~~~
Local Area Connection* 9	_____
Local Area Connection* 8	_____
Adapter for loopback traffic capture	^^ ^.^

1. Select the network adapter with traffic

Learn

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You are running Wireshark 3.2.7 (v3.2.7-0-gfb6522d84a3a). You receive automatic updates.



ARP

```
888 73.928771 Chongqin_... Broadcast ARP 60 Who has 192.168.1.100? Tell 192.168.1.107
889 73.928779 IntelCor_... Chongqin_... ARP 42 192.168.1.100 is at 50:eb:71:bd:fc:f4
```

- ▼ Ethernet II, Src: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 - › Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 - › Source: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98)
Type: ARP (0x0806)
Padding: 00000000000000000000000000000000
- ▼ Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98)
 - Sender IP address: 192.168.1.107
 - Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
 - Target IP address: 192.168.1.100
- ▼ Ethernet II, Src: IntelCor_bd:fc:f4 (50:eb:71:bd:fc:f4), Dst: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98)
 - › Destination: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98)
 - › Source: IntelCor_bd:fc:f4 (50:eb:71:bd:fc:f4)
Type: ARP (0x0806)
- ▼ Address Resolution Protocol (reply)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: reply (2)
 - Sender MAC address: IntelCor_bd:fc:f4 (50:eb:71:bd:fc:f4)
 - Sender IP address: 192.168.1.100
 - Target MAC address: Chongqin_4f:6c:98 (ec:5c:68:4f:6c:98)
 - Target IP address: 192.168.1.107

TCP Handshake: SYN

tcp.stream eq 17						
No.	Time	Source	Destination	Protocol	Length	Info
97	4.69...	192.168.1.3	220.249.46.34	TCP	66	51188 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
98	4.72...	220.249.46.34	192.168.1.3	TCP	66	80 → 51188 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1440 SACK_PERM=1 WS=256
99	4.72...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=1 Ack=1 Win=132352 Len=0
100	4.72...	192.168.1.3	220.249.46.34	HTTP	238	GET /api/toolbox/geturl.php?h=853EA31B28F6A2BAA86434EA7564E605&v=9.5.0.3517&r=0000_sogou_pinyin_94a1 HTTP/1.1
101	4.75...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [ACK] Seq=1 Ack=185 Win=30464 Len=0
102	4.75...	220.249.46.34	192.168.1.3	HTTP	208	HTTP/1.1 200 OK
103	4.75...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [FIN, ACK] Seq=185 Ack=155 Win=132096 Len=0
104	4.79...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [FIN, ACK] Seq=155 Ack=186 Win=30464 Len=0
105	4.79...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=186 Ack=156 Win=132096 Len=0

▼ Transmission Control Protocol, Src Port: 51188, Dst Port: 80, Seq: 0, Len: 0

Source Port: 51188
Destination Port: 80
[Stream index: 17]
[TCP Segment Len: 0]
Sequence number: 0 (relative sequence number)
Sequence number (raw): 2844094833
[Next sequence number: 1 (relative sequence number)]
Acknowledgment number: 0
Acknowledgment number (raw): 0
1000 = Header Length: 32 bytes (8)
> **Flags: 0x002 (SYN)**
Window size value: 64240
[Calculated window size: 64240]
Checksum: 0xc41c [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
> Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted

TCP Handshake: SYN-ACK

tcp.stream eq 17						
No.	Time	Source	Destination	Protocol	Length	Info
97	4.69...	192.168.1.3	220.249.46.34	TCP	66	51188 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
98	4.72...	220.249.46.34	192.168.1.3	TCP	66	80 → 51188 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1440 SACK_PERM=1 WS=256
99	4.72...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=1 Ack=1 Win=132352 Len=0
100	4.72...	192.168.1.3	220.249.46.34	HTTP	238	GET /api/toolbox/geturl.php?h=853EA31B28F6A2BAA86434EA7564E605&v=9.5.0.3517&r=
101	4.75...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [ACK] Seq=1 Ack=185 Win=30464 Len=0
102	4.75...	220.249.46.34	192.168.1.3	HTTP	208	HTTP/1.1 200 OK
103	4.75...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [FIN, ACK] Seq=185 Ack=155 Win=132096 Len=0
104	4.79...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [FIN, ACK] Seq=155 Ack=186 Win=30464 Len=0
105	4.79...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=186 Ack=156 Win=132096 Len=0

✓ Transmission Control Protocol, Src Port: 80, Dst Port: 51188, Seq: 0, Ack: 1, Len: 0

Source Port: 80
Destination Port: 51188
[Stream index: 17]
[TCP Segment Len: 0]
Sequence number: 0 (relative sequence number)
Sequence number (raw): 911860538
[Next sequence number: 1 (relative sequence number)]
Acknowledgment number: 1 (relative ack number)
Acknowledgment number (raw): 2844094834
1000 = Header Length: 32 bytes (8)

➤ Flags: 0x012 (SYN, ACK)
Window size value: 29200
[Calculated window size: 29200]
Checksum: 0x336c [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0

➤ Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale

TCP Connection Termination: FIN

tcp.stream eq 17						
No.	Time	Source	Destination	Protocol	Length	Info
97	4.69...	192.168.1.3	220.249.46.34	TCP	66	51188 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
98	4.72...	220.249.46.34	192.168.1.3	TCP	66	80 → 51188 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1
99	4.72...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=1 Ack=1 Win=132352 Len=0
100	4.72...	192.168.1.3	220.249.46.34	HTTP	238	GET /api/toolbox/geturl.php?h=853EA31B28F6A2BAA86434EA7
101	4.75...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [ACK] Seq=1 Ack=185 Win=30464 Len=0
102	4.75...	220.249.46.34	192.168.1.3	HTTP	208	HTTP/1.1 200 OK
103	4.75...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [FIN, ACK] Seq=185 Ack=155 Win=132096 Len=0
104	4.79...	220.249.46.34	192.168.1.3	TCP	54	80 → 51188 [FIN, ACK] Seq=155 Ack=186 Win=30464 Len=0
105	4.79...	192.168.1.3	220.249.46.34	TCP	54	51188 → 80 [ACK] Seq=186 Ack=156 Win=132096 Len=0

Transmission Control Protocol, Src Port: 51188, Dst Port: 80, Seq: 185, Ack: 155, Len: 0

Source Port: 51188
Destination Port: 80
[Stream index: 17]
[TCP Segment Len: 0]
Sequence number: 185 (relative sequence number)
Sequence number (raw): 2844095018
[Next sequence number: 186 (relative sequence number)]
Acknowledgment number: 155 (relative ack number)
Acknowledgment number (raw): 911860693
0101 = Header Length: 20 bytes (5)

Flags: 0x011 (FIN, ACK)
Window size value: 516
[Calculated window size: 132096]
[Window size scaling factor: 256]

Summary

❑ The OSI model has seven layers, including:

- The application layer: *e.g.*, HTTP, DNS
- The transport layer: TCP ensures reliable transmission.
- The network layer: IP addresses are used for routing over the Internet.
- The data link layer: A MAC address is unique to each device.